Foundations of Data Analytics
Data Fundamentals

1. What is a Database
A database is an organized collection of information that is stored electronically so it can be easily accessed, managed, and updated. Instead of keeping data scattered across paper files or random documents, a database stores everything in a structured way, usually in tables, making it quick to search, sort and retrieve information when needed.

1.1. Why Database are essential in the modern world
Database are essential because they allow organizations to store information safely, keep it accurate and access it istantly. They also support multiple users at the same time, ensure data is backed up, and help protect sensitive information through security controls.
Without Database, Systems like online banking, healthcare systems, mobile apps and business operations would be slow, unreliable and prone to errors.

1.2. Real-world examples of database use.
1.2.1. Banking Systems.
Banks use database to store customers details, accounts balances, transactions history, and loan information. Every time a customer withraws money, checks a balance, or make a payment, the database updates and retrieve information in real time.
1.2.2. Healthcare Systeme.
Hospitals and clinics use database to manage patient records appointment schedules, test results and medication history This ensures doctors and nurses can quickly access accurate patient information, which improves efficiency and care.

2. What is a Data Warehouse.

A data Warehouse is a Centralized system used to store Large amounts of data Collected from different Sources over time. It's main purpose is to Support reporting, analysis and decision making rather than daily operations. The data in a data warehouse is usually cleaned, organized, and structured in a way that makes it easy to analyse trends and patterns.

2.1. How does a Data Warehouse differ from a regular database

A database is designed to handle day to day transaction, While a data warehouse is designed to analyse data for trends and patterns.

2.1.2. Why would a Company use a data warehouse.

A Company uses a data warehouse for it's purpose on Supporting analysis, reportings and strategic decision making. Also because of large Volumes of data Storage that changes less often. As well as performance focus because data Warehouse are optimized for Complex queries, Sumaries and data analysis. Again because of it ability to Combine data from many Systems (e.g. Finance, Sales, HR, and Customer Systems) into one place.

3. What is a data Lake.

Data Lake is an extensible Storage that stores all types of data i.e. Structured, semi-structured and unstructured data in it's raw format.

3.1. Explain the Concept and describe how a data lake differs from data Warehouse.

In a data lake, data is structured after it is stored (schema on read) where as in a data warehouse, data is Structured before storage (schema-on-write)

3.2. When would a company/business prefer a data lake over a data warehouse

A busines would prefer a data lake over a data warehouse when it needs flexibility, scale and room to explore data that is not yet well defined.
A. data warehouse works better when a business already knows exactly what questions it wants to ask. The data is cleaned, structured and shaped before it is sorted, which makes ~~r~~ reporting and dashboards fast and reliable. It is ideal for regular business reporting and decision-making.
Where-as a data lake on the other hand is better when the business is still figuring things out, and dealing with many different types of data like emails, documents, sensor data, logs, images and social media data in its raw format.

4. What is data Mart

A data mart is a smaller, focused subset of data that is designed to meet the needs of a specific department or business function instead of containing data for the entire organisation.

4.1. How does a data mart relates to a data warehouse

A data warehouse stores large volume of intergrated, organisation data whereas a data mart usually draws its data from a data warehouse and contains only the portion needed for specific purpose. In other words, the data warehouse is central source, and data marts are specialised views of that data tailored to departments or teams. In some cases, a data mart can exist on its own but most commonly it is built from a data warehouse.

4.2. Who typically use data marts

Data marts are typically used by:
- Department managers.
- Business analysts.
- Finance, HR, Sales teams
- Operational decision makers

5. The difference between Structured, Semi Structured and unstructured Data.

5.1 Structured data is a highly organised and follows fixed Format. It fits neatly into rows and Columns and is easy to store, search and analyse using database.

5.2. Semi Structured data does not follow a strict table Format but still Contains tags, labels or markers that gives it some Structure. This makes it more flexible than Structured data, but still easier to analyse than unstructured data.

5.3. Unstructured data has no predefined structure and does not follow a specific format. It is usually Created by people and it is more difficult to analyse without Specialised tools.

Examples

Structured data. e.g. (Customer ID, Name, Date of birth and account balance)

Semi Structured data e.g. (A JSON or XML file that stores details, where information is labelled but not stored in rows and Colum[ns])

Unstructured data e.g. (Emails, Word documents, PDF Files, images, Videos and Social medi[a] posts.

6. What is Data Modeling.

Data Modeling is the process of designing how data will be structured, stored within a System. It defines What data is needed, how different pieces of data Connect to each other and how the data will be used. It acts as a blueprint for building databases and data Systems.

6.1 Why is data modeling important when designing a System that works with data

Data Modeling is important because it ensures that data is Accurate and Consistent, Well organised, Scalable Efficient and that it alignes with business needs.

6.2. Two types of data models.

Conceptual data Model
Logical data Model

7. What is data mining.

Data mining is the process of analysing large amounts of data to find hidden patterns, trends, and useful info. It helps turn raw data into knowledge that business Can use to make decisions.

7.1. How is data mining used in businesses.

Data mining is the process of analysing large amounts of data and businesses use it understand Customer behaviour, improve decision making, predicts future trends, Detects risks or fraud, and increase profits and efficiency.

7.2. Two real world examples of data mining in action

Retail and Online Shopping.
Banking and Fraud detection

8. What is a Dabase Management System (DBMS)

A database management System is a Software that allows users to Create, Store, manage and retrieve data in database. It acts as an interface between users or applications and the database, making sure data is organised, secure, and easy to access.

8.1 Four popular descriptions of Database Management System platform.

8.1.1. MySQL is an open source relational DBMS that stores data in tables using rows and columns. It is known for being fast and easy to use.

Suitable Use Case:

Used in web applications such as online stores and websites (e.g. storing user account, products and others.

8.1.2 PostgreSQL is an advanced open source relation DBMS that supports complex queries and large datasets. It is reliable and flexible

Suitable Use Case:

Used for enterprise system and analytics, such as financial systems or applications that require complex data relationships and high data integrity.

8.1.3. Oracle Database is a commercial enterprise level relational DBMS designed to handle very large, mission critical systems. It offers strong security and performance

Suitable Use Case:

Used in banks and large organisations to manage transactions heavy system like payroll, customer records and financial operations.

8.1.4. MongoDB is a NoSQL DBMS that stores data in flexible, JSON like documents instead of tables. It handles unstructured and semi-structured data well.

Suitable Use Case

Used in mobile apps and real time application, such as social media platforms or product catalog systems where data structure changes often.

9. What is Extract, Transform Load ETL

ETL is a data integration process used to collect data from different sources, clean and organise it and then store it in a central system such as a data warehouse or data lake for analysis. ETL forms the foundation of data analytics and business intelligence because it prepares raw data so it can be reliably analysed.

Three steps of ETL

9.1. Extract - This step involve collecting raw data from multiple sources systems such as database, spreadsheet, applications and websites. The data is taken as-is and stored temporarily for processing.

9.2. Transform - In this step the extracted data is cleaned and converted into a consistent and usable format. Common transformation ~~tasks~~ tasks includes, removing duplicates, fixing errors, standardising formats (dates, currencies) and combining data from different sources. This ensures the data is accurate and ready for analysis.

9.3. Load - The final step is loading the transformed data into a targeted system such as a data warehouse, data mart or data lake. Once loaded, the data is available for reporting, dashboards and analytics.

9.4. Why ETL is important in Data Analytics.

ETL is important because it improves data quality and consistency, it combines data from multiple sources into one view. It ensures reliable analysis and reporting, as well as it supports data driven decision making. Without ETL, organisations would analyse incomplete or inconsistent data resulting to poor decisions.

10. Common Data Storage Formats Used in Alalytics.

10.1. CSV – Comma Separated Values.

CSV is a simple text based format where data is stored in rows and columns, separated by commas. It is easy to read and supported by most tools. It can be used on small to medium datasets, Quick data sharing as well as simple analysis using Excel or basic tools.

10.2. JSON – JavaScript Object Notation.

JSON stores data as key value pairs and supports nested structures. It is commonly used for semi-structured data. It can be used on semi-structured data, on data from API's or web applications and also when flexibility is needed.

10.3. Parquet

Parquet is a column based, compressed file format designed for big data analytics. It stores data by columns instead of rows. It can be used on large datasets, Datawarehous and data lakes and it is faster ~~on~~ on analytics and reporting. Highly efficient for analytics and Cloud platforms.

10.4. AVRO

Avro is a row-based, binary data format that include a schema. It is commonly used in data streaming and big data pipeline. It can be used on streaming data, Data intergration and pipelines, also when schema evolution is required. It is good for data ingestion and system communication.

Format	Data Type	Best Used for.
CSV	Structured	Small datasets & Spreadsheet
JSON	Semi-Structructured	API's, Web data.
Parquet	Structured	Big data analytics
AVRO	Structured	Data Streaming & pipelines

Section B, Artificial Intelligence & Machine learning.

11. What is Artificial Intelligence (AI)

It is the ability of a computer or a machine to perform tasks that normally require human intelligence. AI is designed to mimic certain aspects of human thinking and behaviour.

11.1. Difference between Narrow AI (weak AI) and General AI (strong AI)

Narrow AI is a type of AI that is designed to perform one specific task or limited set of tasks. It works very well within its defined area but cannot think or act beyond that task. Focused on a single function and cannot learn or reason outside its programming.

Example of Narrow AI: A recommendation system on netflix that suggests movies based on your viewing history.

General AI is a theorical form of AI that would have human like intelligence. It understands, learn and perform any intellectual task that a human can, across different areas. It can reason and adapt to new situations, learn across multiple tasks and does not yet exist in reality.

Example of General AI: A robot that can learn new skills, solve unfamiliar problems, and perform different jobs like a human being. (This is still theoretical).

Type of AI	Task Specific Intelligence	Netflix Recommendations
Narrow AI	Task Specification	Netflix Recommendations.
General AI	Human level - Flexible Intelligence.	Human like intelligent Robot.

12. What is Machine learning

Machine learning is a type of artificial intelligence that enables computers from data on improve their performance over time without being explicitly programmed. Instead of following rules (hard coded), machine learning systems identifies patterns in data, and use those patterns to make predictions and decisions.

12.1. How does Machine learning differ from traditional programms
In traditional programming a human writes clear rules and instructions, and the Computer follows those rules to produce an output.
In Machine learning the Computer is given data and expected outcomes, and it learns the rules on its own.
12.2. Describe the three main types of machine learning
Supervised learning - is a type of machine learning where the model is trained using labelled data, learns a direct relationship between inputs and known outcomes.
Unsupervised Learning - Works with unlabelled data, meaning the model is not given predefined outcomes. It identifies patterns, groupings or anomalies
Reinforcement learning - is a type of machine where agents learns through interactions with environment. The agent takes actions, receives feedback in the form rewards or penalties, and learns over time which actions lead to the best long outcomes.
13. What is Deep Learning.
Deep learning is a subset of machines learning that uses multi Layered artificial neural networks to automatically learn patterns from large amounts of data.
13.1. How does it relate to machine learning and artifical intelligence.
Deep learning fits within a clear hierachy of
Artificial Intelligence - which is broad field focused on making machine perform tasks that require human intelligence.
Machine learning - Which is a Subset of AI Where systems learn data rather than fixed rules.
Deep learning - Which is a specialized type of ML that uses neural networks with many layers.

14. What is Natural language Processing (NLP)
NLP is a branch of artificial intelligence that enables Computers to understand, interpret and respond to human language in a useful way.
14.1 Real world application of NLP used in everyday life
1. Voice assistants (Siri, Alexa, and google assistant)
These use NLP to understand spoken language, Convert speech into text.
2. Email spam phishing.
This service use NLP to analyse the Content of emails and identify spam or phishing messages, based language patterns and key words.
3. Text prediction and Autocorrect
NLP is used in smartphones and messaging apps to predict the next word, Correct spelling and improve typing Speed by understanding sentence Context
15. What is a large language model (LLM)
LLM is a type of deep learning model designed to understand generate and work with human language. It is trained on massive amounts of data (such as books, websites etc) and learn language pattern by predicting the next word in Sentence.
15.1. How is it different from a standard Machine learning Model.
This is usually trained to perform one specific task, such as predicting house prices or classifying emails as spam. It relies on carefully prepared features and works on narrow database. It is trained on huge, diverse text database, Can perform many language related tasks, understand Context grammar and meaning, Can generate human like responses.

15.2. Two well known LLMs and explain what they used for.

- GPT (Generative Pre-trained Transformer)
It is used for Chatbots and virtual assistants e.g. ChatGPT
For Writing emails, essays and report.
For Assisting with programming and debugging code.
- BERT (Bidirectional Encoder Representations from Transformers
It is used for improving google Search
For understanding Search queries
For Sentiment analysis and text classification.

16. What is generative AI

Generative AI is a branch of artficial intelligence that focuses on Creating new Content rather than only analysing or Classifying existing data. It Can generate text, images Audio, video or Code that resembles Content created by humans.

16.1. How generative AI works (High level Explanation)

At a high level, generative AI works as follows:
16.1.1. Training on large amounts data i.e. text, images or Sound
16.1.2. It learns learns patterns, Structure, and relationships in data.
16.1.3. It predicts and generates new Content based on what it has learned.

16.2. Examples and tools of generative AI

- ChatGPT it generates human like text based on prompt
- Microsoft CoPilot/Github Co-pilot assists programmers by generating Code in real tim.
- DALL•E it Creates images from text description.

17. What is a Training dataset

A training dataset a Collection of data used to teach an AI or machine learning model How to perform a task.

17.1. Why is the quality of training data so important when building an AI model

It is because the quality of data (training data) determines How well the AI model will perform. AI Systems learn exactly what the data Shows them.

17.2 What Could go wrong if the training data is biased or incorrect.

If the training data is biased, incomplete or incorrect, Several Serious problems Can occur

Summary Table.

Aspect	Impact
Good training data	Accurate, Fair AI Model
Poor or biased data	Errors, Unfair decisions
Incorrect labels	Unreliable Predictions
Lack of diversity	Discrimination and bias

18. What is Computer Vision.

Computer Vision is a branch of artificial intelligence that enables machines to analyse and understand images and videos in a meaningful way.

18.1. How do machines see and interpret images.

Machines do not see images like humans do. Instead they interpret images as numbers (pixel) and analyse those numbers using AI models.

18.2. Two real word use Cases computer vision.

-> Facial recognition and phone unlocking.

-> Medical imaging and health Care (e.g. X rays)

Section C: API's Tools and Developer Concepts.

19. What is an API (Application Programming interface)

An API is a set of rules that allows different Software systems to Communicate with each other. It defined how one application can request data or Service from application and how that application should respond.

19.1. Explain real-world analogy of API Usage.

In real world simple analogy, will think of an API as Waiter in a restaurant. - You (the use/app) look at the menu and place and order.
- The waiter (API) takes your order to the kitchen
- The kitchen (system/database) prepares the food
- The waiter (API) brings the food back to you.

You never went into the kitchen yourself, and you dont need to know how the food was Cooked/prepared. You just used the waiter (API) as the interface.

An application makes a request through API, then it talks to another System, Get the results and returns the results. (data / Service).

19.2. Why are API's important in mordern Software and data System

Enables System intergrations

Supports data sharing and automation.

Improves scalability and flexibility.

Power modern digital experience.

20. What is an API key.

An API key is a Unique Secrete Code used to identify and authenticate an application when it makes a request to an API.

An API key acts like password for a software allowing an API to recognize who is making the request and whether they are allowed to access the service or data

20.1. Purpose of API key

API keys are used to identify, Authorize access, Track usage and prevent abuse on applications.

20.2 How should an API key be stored an protected.

Best practices is to store API keys securly on the server side not in public code.
Use environmental variables or secure secrete management tools
Not to be hard-coded into source code or share them publicly.
Rotate keys regularly to reduce long term risk.
Permmission or scopes to be used to limit its usage.

20.3. What could happen if an API key is exposed publicly

1. Unauthorized access
2. Financial Loss
3. Data breaches and security incidents.
4. Loss of trust and compliance issues.

21. What is a REST API

A rest API is an API that follows the REST architectured style to allow systems to communicate over the web using standarad HTTP protocols. These are commonly used to let client application interact with servers by requesting or modifying data in a simple predictable way.
REST stands for Representational State Transfer

2.1.1. Main HTTP Method used in REST API's.

2.1.1.1. GET – retrieve data from the server.
2.1.1.2. Post – sends data to the server to create a new resource.
2.1.1.3. PUT – Updates or replace an existing resource.
2.1.1.4. DELETE – Removes a resource from the server.

2.2 What is JSON (JavaScript object Notation)

It is a light weight text based data format used to represent and exchange structured data. It stored data in in a key value pairs and arrays, making it easy for both humans to read and machines to parse.

2.2.1. Why is JSON commonly used when working with API and data exchange

1. It is easy to read and write.
2. Lightweight and fast
3. Works across different systems
4. Fits naturally with API

2.2.2. A Example of a JSON

```
{
    "Name": "Angel",
    "Channel": "ATR Speaks"
    "Topics": ["Tech", "cloud"]
    " Views": 1500
}
```

Citation from youtube Channel @artspeaks.

2.3. What is prompt engineering.

It is a practice of designing clear, specific and well structured prompt to guide an AI model towards more accurate, relevant and useful responses. A prompt is a instruction or context.

2.3.1. Importance of prompts when working with AI tools

because AI models do exactly what they are asked no more, no less.

Better prompts produce higher quality outputs

They are effecient and reliable in professional use.

23.2. Two examples of a poorly writen prompt and one improved version of each.

Poorly Written Prompt	Improved Written Prompt.
"Explain AI"	explain artificial inteligence in simple terms for a First Student. Use short definitions.
Write an email about the meeting	Write a polite, proffessional follow up email to team summerising key decisions from yesterday's meeting

24. What is a Map Function

Map function is a programming concept used to apply the same operation to every item in collection of data, and return a new collection of results. Istead of writing a loop and updating values one by one, the MAP function performs the transformation autometically and consistently.

24.1. What does MAP function do in programming and data processing

At a high level Map takes a function (What you want to do) It takes a collection of data (list, array, iterable) It applies the function to each element and returns a new collection with the transformed values.

24.2. Example of MAP Function in Python

```
numbers = (1, 2, 3, 4 )
squared/doubled = map (lambda x: x**2, numbers )
results = List (squared )
```

Citation from Youtube Channel @learnTechnology.

25. What is Cloud Computing.

Cloud computing is the delivery of Computing services over the internet instead of using physical Computers or servers at your own location. These include servers, storage, database application and software.

25.1. Three main main cloud services models

1. IaaS - provides basic Computing infrastructure such as virtual servers, storage and Networking. It is an Infrastructure as a Service.

Example of IaaS.
→ Amazon EC2 – Virtual servers that users can configure on manage themselves.
2. PaaS – Platform as a Service.
It provides a ready to use platform that allows developers to build, test and deploy applications without worrying about the underlying infrastructure.
Example of PaaS
→ Google App Engine – a platform for developing and running application in the Cloud.
3. SaaS – Software as a Service.
It delivers fully functional Software over the internet. Users do not manage infrastructure or platforms, they simply use the Software through a web browser or app.
Example of SaaS
→ Microsoft outlook (office 365) email, and productivity Software accessed online.
26. What a version Control System.
It is a tool that Helps people track changes made to files over time. It allows users to Safe different versions of their work, go back to earlier versions if Something goes wrong.
26.1 What is a Git and why it used by data analysts and developers
Git is a popular version Control System used to manage and track Changes in files, especially Source Code.
It is used by data analysts and developers because
- it helps track changes in Code and analysis files.
- it Supports teamwork and Collaboration.
- it makes it easier to experiment without breaking the main project.
- it allows users to recover previous versions if errors occur.

26.2. Brief explanation of what a "Commit" and "branch" are.

1. A Commit is a saved snapshot of changes made to files at a specific point in time, it records what was changed and when.
2. A branch is a separate copy of the project used to work on new features or changes without affecting the main version.

27. What is Jupyter Notebook

A Jupyter notebook is an open source, web based interactive computing environment that allows users to create and share documents containing: live executable code
: Text and explanations
: Mathematical equations
: Data visualization.

27.1. Programming language used with Jupyter.

Python is the most commonly used language used in Jupyter. in the use of its libraries of Pandas, NumPy, Matplotib etc.

27.2. Why is Jupyter Popular amongst data analysts.

Jupyter notebook is popular in data analysis and data science because they support the entire analytical work-flow.

28. What is Python

Phython is a high level, open source, interpreted programming language know for it simple readable syntax and versatility.
It is used and considered the most popular programming language because of ease of use and readability, Extensive library ecosystem, Strong support for AI and Machine learning as well as large community and industry adoption.

28.1. Four Python libraries.

1. Pandas
2 NumPy
3 Matplotlib
4. Scikit-learn

Section D: Data Ethics, Governance and Careers.

29. What is data privacy.

Data privacy refers to the right of individuals to control how their personal information is collected, used, stored and shared, by organisations.

Data privacy is important because it protects both individuals and organisations

General Data Protection Regulations (GDPR) is a comprehensive data protection law introduced by the European Union (EU). to regulate how personal data is processed, stored and shared.

30. What is data Ethics.

Data ethics refers to the moral principles and standard that guide how data and AI systems are collected, used, shared and managed.

30.1. Three unethical examples of use of data.

1. Algorithmic bias in AI decision making.

Some AI systems used for decision making such as hiring, credit scoring, health care prioritisation or policing produce biased outcomes because they are trained on historical data.

-Why is it unethical- Decision become harder to challenge because the AI process is often opaque.

2. Mass surveillance and facial recognition without consent

This is sometimes used to collect and analyse people's biometrics data without their knowledge or consent.

It undermines the right to privacy and personal authority.

3. Creation and use of AI generated deepfakes.

Deepfakes create realistic but fake images, videos, or audios of people saying or doing things that never happened. resulting in reputation damage.

3.1. What is data governance.
Data governance is the way an organisation takes responsibility for its data. Set out rules, processes and standards that ensures data accuracy.

3.1.1 Key Components of a good data governance.

1. Clear role and standards
2. Policies and standards.
3. Data quality management
4. Data security and privacy controls.
5. Data stewardship and lifecycle management
6. Communication, training and awareness.

3.2. What is Bias in AI

Bias in AI refers to a situation where an artificial intelligence system produce unfair, inaccurate, or prejudiced outcome for certain individuals or groups. When it is systematically favors or disadvantage people.
Criminal risk assessment tool used in USA. This was designed to help courts predict how likely a defendant was to reoffend (commit another crime). Investigations later showed that the system incorrectly rated black defendants as high risk more than white defendants, even when both groups had similar backgrounds and offense history.
- Resulting in unfair treatment
- Serious world consequences
- Loss of trust
- Reinforcement of inequality.

3.3.

3.3. key differences between data AI related roles

Roles	Main Focus / Role	key Skills	Common tools & Technology.
Data Analyst	Analysis existing data to find insights and explain what has happened and why. Focuses on reporting and supporting business decisions.	Data Cleaning, data analysis, Statistics, data visualization, Communication	Excel, SQL, power BI, Python, Tableau, Pandas, Google analytics
Data Scientist	Uses data to build models and make predictions about future outcomes. Works on advanced analytics, machine learning and experimentation	Statistics, Machine learning Programming, Modeling Problem-Solving	Python, R, SQL Jupyter notebook, Tensorflow, Scikit-learn Spark
Data Engineer	Builds and maintains the System that collect, Store, and process large amounts of data. Ensure data is reliable and accesible.	Data pipelines, database ETL, Cloud platforms System design	SQL, Python, Spark, Hadoop Azure, Airflow Databricks AWS
AI Engineer	Design, trains, deploys and maintain AI and machine learning models in real world application Focuses on production-ready AI Systems.	Machine learning, deep learning, Software engineering, Model deployment	Python, TensorFlow PyTorch, API's Docker, Kubernetes Cloud AI Services

34. What is business intelligence.

This refers to the processes, technologies and tools used to Collect analyses and present business data.

3.4.1. Two popular BI tools

→ Power BI is a business intelligence tools developed by Microsoft. It helps busesses connect to multiple data Sources, analyse data and Create interactive dashboards and reports

34.2. Tableau - is a widely used BI and data visualisation tool that helps users see and understand data through interactive visuals. It turns complex data into clear charts graphs and dashboards, Help users explore data, spot patterns and identify trends, also support - faster decision making by making data easy to interpret across teams.

Section E: Personal Reflection

35. The question i found surprising is question 33. on the difference of designations in AI. I had a mind that it was a narrow field with no versatile professions.

36. Section C on API's Tools & developer Concepts. I used Co-Pilot to assist me with this section.

3.7. How i see data and AI affecting my future career.
- Based on where i am right ~~now~~, working as a Personal Assistant at an insurance company while actively actively learning about data analytics, AI and related tools. I see data and AI playing a very direct and practical role in both my future career growth and my every-day work life.

Impact on my current and everyday work life

I already see how AI can improve effeciency and accuracy for instance. - AI can help me work faster when preparing reports, SOP, or policies.

- Data tools can help me spot errors, gaps or trends that might be missed when working manually. I dont see it as threat but as a tool that will help me grow professionally.

Sources I used to complete the assignment are as follows.

Co-pilot

You-tube - Channel @learnTechnology.
- Channel @artspeak.